

## A. DETAILED PROOFS

### A.1. Proof of Theorem 1

*Proof.* Let  $d \geq 3$ , and consider two distinct prototype locations  $\mathbf{p}_1, \mathbf{p}_2 \in \mathbb{R}^d$ . We construct probabilistic soft labels whose class-specific affine scores are uniquely dominant on  $K$  non-empty intervals of the distance ratio.

Define the normalization constant  $S = \sum_{\ell=1}^{K-1} \ell^2$ . For  $k = 1, \dots, K$ , define

$$y_{1k} = \frac{\sum_{\ell=k}^{K-1} \ell}{S}, \quad y_{2k} = y_{1, K+1-k}, \quad (1)$$

where an empty sum is defined as zero. In particular,  $y_{1K} = 0$ , which is allowed by the probabilistic soft-label definition.

We first verify that  $\mathbf{y}_1$  and  $\mathbf{y}_2$  are valid probability vectors. Every entry is non-negative, and

$$\sum_{k=1}^K y_{1k} = \frac{1}{S} \sum_{k=1}^K \sum_{\ell=k}^{K-1} \ell = \frac{1}{S} \sum_{\ell=1}^{K-1} \sum_{k=1}^{\ell} \ell = 1. \quad (2)$$

Since  $\mathbf{y}_2$  is the reverse ordering of  $\mathbf{y}_1$ , its entries are also non-negative and sum to one.

For  $k = 1, \dots, K-1$ , the adjacent differences satisfy  $y_{1k} - y_{1, k+1} = k/S$  and  $y_{2, k+1} - y_{2k} = (K-k)/S$ . Using Equation (4), the difference between the affine scores of classes  $k$  and  $k+1$  has the same sign as  $k\rho(\mathbf{z}) - (K-k)$ . Therefore,

$$g_k(\mathbf{z}) > g_{k+1}(\mathbf{z}) \iff \rho(\mathbf{z}) > \frac{K-k}{k}, \quad k = 1, \dots, K-1. \quad (3)$$

These adjacent boundaries are strictly decreasing because, for  $k = 1, \dots, K-2$ ,  $\frac{K-k}{k} - \frac{K-k-1}{k+1} = \frac{K}{k(k+1)} > 0$ . Hence,

$$K-1 > \frac{K-2}{2} > \dots > \frac{1}{K-1} > 0. \quad (4)$$

We next identify the uniquely dominant class on each interval. If  $\rho(\mathbf{z}) > K-1$ , then  $g_1(\mathbf{z}) > g_2(\mathbf{z}) > \dots > g_K(\mathbf{z})$ , so class 1 is the unique maximizer. For any  $k \in \{2, \dots, K-1\}$ , consider

$$\frac{K-k}{k} < \rho(\mathbf{z}) < \frac{K-k+1}{k-1}. \quad (5)$$

For every  $j < k$ , the adjacent boundaries imply  $g_j(\mathbf{z}) < g_{j+1}(\mathbf{z}) < \dots < g_k(\mathbf{z})$ , whereas for every  $j > k$ ,  $g_k(\mathbf{z}) > g_{k+1}(\mathbf{z}) > \dots > g_j(\mathbf{z})$ . Thus, class  $k$  is the unique maximizer throughout the interval in Equation (5). Finally, if  $0 < \rho(\mathbf{z}) < 1/(K-1)$ , then  $g_1(\mathbf{z}) < g_2(\mathbf{z}) < \dots < g_K(\mathbf{z})$ , so class  $K$  is the unique maximizer.

The resulting decision rule is therefore

$$\arg \max_{k \in \{1, \dots, K\}} g_k(\mathbf{z}) = \begin{cases} 1, & \rho(\mathbf{z}) > K-1, \\ k, & \frac{K-k}{k} < \rho(\mathbf{z}) < \frac{K-k+1}{k-1}, \quad 2 \leq k \leq K-1, \\ K, & 0 < \rho(\mathbf{z}) < \frac{1}{K-1}. \end{cases} \quad (6)$$

At each omitted boundary  $\rho(\mathbf{z}) = (K-k)/k$ , classes  $k$  and  $k+1$  have equal scores.

It remains to show that every strict decision region is non-empty and open. Consider points on the open segment connecting the two prototypes,  $\mathbf{z}_t = (1-t)\mathbf{p}_1 + t\mathbf{p}_2$  with  $t \in (0, 1)$ . For such points,  $\rho(\mathbf{z}_t) = \|\mathbf{z}_t - \mathbf{p}_2\|_2 / \|\mathbf{z}_t - \mathbf{p}_1\|_2 = (1-t)/t$ . The map  $t \mapsto (1-t)/t$  is continuous from  $(0, 1)$  onto  $(0, \infty)$ . Therefore, every interval in Equation (6) is attained by a non-empty subset of this segment.

For each class  $k$ , let  $I_k$  denote its corresponding open interval in Equation (6), and define

$$\mathcal{R}_k = \left\{ \mathbf{z} \in \mathbb{R}^d \setminus \{\mathbf{p}_1, \mathbf{p}_2\} : \rho(\mathbf{z}) \in I_k \right\}. \quad (7)$$

The function  $\rho$  is continuous on  $\mathbb{R}^d \setminus \{\mathbf{p}_1, \mathbf{p}_2\}$ . Since  $I_k$  is open,  $\mathcal{R}_k = \rho^{-1}(I_k)$  is open, and the segment construction above shows that it is non-empty. Hence, for every  $k \in \{1, \dots, K\}$ ,  $g_k(\mathbf{z}) > g_j(\mathbf{z})$  for all  $\mathbf{z} \in \mathcal{R}_k$  and all  $j \neq k$ . Thus, every candidate class is realized as the unique maximizer on a non-empty open decision region induced by two prototype locations.  $\square$

## A.2. Proof of Theorem 2

*Proof.* Throughout,  $\mathbf{z} \sim \text{Uniform}([-3, 3]^d)$ ,  $\mathbf{p}_1 = -\mathbf{e}_1$ ,  $\mathbf{p}_2 = +\mathbf{e}_1$ , and  $R^2 = \sum_{i=2}^d z_i^2$ . Then

$$\|\mathbf{z} - \mathbf{p}_1\|_2^2 = (z_1 + 1)^2 + R^2, \quad \|\mathbf{z} - \mathbf{p}_2\|_2^2 = (z_1 - 1)^2 + R^2. \quad (8)$$

The denominator in  $\rho(\mathbf{z})$  is positive almost surely under the continuous sampling distribution.

Subtracting the squared distances in Equation (8) gives  $\|\mathbf{z} - \mathbf{p}_2\|_2^2 - \|\mathbf{z} - \mathbf{p}_1\|_2^2 = -4z_1$ . Factoring the left-hand side and dividing by the positive terms yields

$$|\rho(\mathbf{z}) - 1| = \frac{4|z_1|}{(\|\mathbf{z} - \mathbf{p}_2\|_2 + \|\mathbf{z} - \mathbf{p}_1\|_2) \|\mathbf{z} - \mathbf{p}_1\|_2}. \quad (9)$$

Since  $\|\mathbf{z} - \mathbf{p}_2\|_2 + \|\mathbf{z} - \mathbf{p}_1\|_2 \geq \|\mathbf{z} - \mathbf{p}_1\|_2$  and  $|z_1| \leq 3$ ,

$$|\rho(\mathbf{z}) - 1| \leq \frac{12}{\|\mathbf{z} - \mathbf{p}_1\|_2^2}. \quad (10)$$

Conversely,  $\|\mathbf{z} - \mathbf{p}_1\|_2 \leq \|\mathbf{z}\|_2 + 1$  and  $\|\mathbf{z} - \mathbf{p}_2\|_2 + \|\mathbf{z} - \mathbf{p}_1\|_2 \leq 2(\|\mathbf{z}\|_2 + 1)$ , so

$$|\rho(\mathbf{z}) - 1| \geq \frac{2|z_1|}{(\|\mathbf{z}\|_2 + 1)^2}. \quad (11)$$

Let  $m = d-1$  and write  $R^2 = \sum_{i=1}^m w_i^2$ , where the  $w_i$  are independent and uniform on  $[-3, 3]$ . Define  $S^2 = \|\mathbf{z} - \mathbf{p}_1\|_2^2 = (z_1 + 1)^2 + R^2$ . Since  $\mathbb{E}[w_i^2] = 3$  and  $0 \leq w_i^2 \leq 9$ , Hoeffding's inequality gives, for an absolute constant  $c > 0$ ,

$$\mathbb{P}\left[S^2 \leq \frac{3m}{2}\right] \leq \mathbb{P}\left[R^2 \leq \frac{3m}{2}\right] \leq e^{-cm}. \quad (12)$$

For sufficiently small  $s > 0$ , the event  $\{S^2 \leq s\}$  lies inside a ball of radius  $\sqrt{s}$  centered at  $\mathbf{p}_1$ , and hence

$$\mathbb{P}[S^2 \leq s] \leq Cs^{d/2}, \quad (13)$$

where  $C$  is uniformly bounded in  $d$ .

*Upper bound.* Using the tail-integral representation of the inverse moment and splitting the integral at 1 and  $3m/2$ ,

$$\begin{aligned} \mathbb{E}\left[\frac{1}{S^2}\right] &= \int_0^\infty \mathbb{P}[S^2 < s] \frac{ds}{s^2} \\ &\leq \int_0^1 Cs^{d/2-2} ds + \int_1^{3m/2} e^{-cm} \frac{ds}{s^2} + \int_{3m/2}^\infty \frac{ds}{s^2} \\ &= O\left(\frac{1}{d}\right). \end{aligned} \quad (14)$$

The first integral is finite for  $d > 2$ , which gives the condition  $d \geq 3$ . Combining Equations (10) and (14),

$$\mathbb{E}|\rho(\mathbf{z}) - 1| = O\left(\frac{1}{d}\right). \quad (15)$$

*Lower bound.* Let  $\mathcal{E} = \{R^2 \leq 6m\}$ . Hoeffding's inequality gives  $\mathbb{P}[\mathcal{E}^c] \leq e^{-2m/9}$ , so  $\mathbb{P}[\mathcal{E}] \rightarrow 1$ . On  $\mathcal{E}$ ,  $\|\mathbf{z}\|_2^2 \leq 9 + 6m$  and  $(\|\mathbf{z}\|_2 + 1)^2 \leq 12m + 20$ . Because  $z_1$  is independent of  $R^2$  and  $\mathbb{E}|z_1| = 3/2$ , Equation (11) yields

$$\begin{aligned} \mathbb{E}|\rho(\mathbf{z}) - 1| &\geq \frac{2\mathbb{E}[|z_1|\mathbf{1}_{\mathcal{E}}]}{12m + 20} \\ &= \Omega\left(\frac{1}{m}\right) = \Omega\left(\frac{1}{d}\right). \end{aligned} \quad (16)$$

Combining Equations (15) and (16) gives

$$\mathbb{E}[|\rho(\mathbf{z}) - 1|] = \Theta\left(\frac{1}{d}\right), \quad (17)$$

which proves Theorem 2.  $\square$

*Remark 1* (Numerical behavior of the scaled deviation). The numerical study additionally evaluates  $d\mathbb{E}[|\rho(\mathbf{z}) - 1|]$ . Across the tested dimensions, the scaled quantity remains close to a constant near 1, which is consistent with the inverse-dimensional rate established in Theorem 2.

### A.3. Proof of Corollary 1

*Proof.* By definition,  $\rho(\mathbf{z}) \in \mathcal{A}_\delta$  implies  $|\rho(\mathbf{z}) - 1| \geq \delta$ . Applying Markov's inequality to the non-negative random variable  $|\rho(\mathbf{z}) - 1|$  gives

$$\begin{aligned} \mathbb{P}[\rho(\mathbf{z}) \in \mathcal{A}_\delta] &\leq \mathbb{P}[|\rho(\mathbf{z}) - 1| \geq \delta] \\ &\leq \frac{\mathbb{E}[|\rho(\mathbf{z}) - 1|]}{\delta} \\ &= O\left(\frac{1}{\delta d}\right), \end{aligned} \tag{18}$$

where the final step follows from Theorem 2. Thus, probability mass assigned to distance-ratio values separated from 1 vanishes as the dimension increases.  $\square$

## B. IMPLEMENTATION DETAILS FOR THE THEORETICAL NUMERICAL STUDY

The numerical study in Section 4.2 uses the same geometric setting as Theorem 2, with  $\mathbf{p}_1 = -\mathbf{e}_1$ ,  $\mathbf{p}_2 = +\mathbf{e}_1$ , and  $\mathbf{z} \sim \text{Uniform}([-3, 3]^d)$ . We evaluate  $K \in \{2, 3, 4, 5, 6, 8, 10, 20\}$  and  $d \in \{3, 5, 10, 20, 50\}$ , drawing  $4 \times 10^5$  independent uniform samples for each configuration.

The numerical construction uses the strictly increasing geometric schedule  $t_i = 0.25 \cdot 1.10^{i-1}$  for  $i = 1, \dots, K-1$  to control the ordering and relative spacing of adjacent affine-score intersections. The implementation first constructs two affine coefficient vectors from this ordered schedule and then normalizes the two vectors separately to obtain the probabilistic soft labels  $\mathbf{y}_1$  and  $\mathbf{y}_2$ . Separate normalization rescales all adjacent intersections by the same positive factor, so their strict ordering and the number of non-empty winning intervals are preserved. For a fixed  $K$ , the same normalized soft labels are used across all tested dimensions.

For each sampled point  $\mathbf{z}$ , define  $d_1 = \|\mathbf{z} - \mathbf{p}_1\|_2$ ,  $d_2 = \|\mathbf{z} - \mathbf{p}_2\|_2$ , and  $\rho = d_2/d_1$ . Predictions are computed independently from the original distance-based score

$$g_k(\mathbf{z}) = \frac{y_{1k}}{d_1} + \frac{y_{2k}}{d_2}, \tag{19}$$

and the affine distance-ratio score  $s_k(\mathbf{z}) = y_{1k}\rho + y_{2k}$ . The two formulations are checked for identical predicted class indices on the sampled points. The recovered class count  $R$  is the number of distinct class indices observed under uniform sampling.

At the reference dimension  $d = 3$ , the volume of class  $k$  is estimated by

$$\hat{V}_k = 6^3 \frac{n_k}{N}, \tag{20}$$

where  $n_k$  is the number of uniformly sampled points assigned to class  $k$  and  $N$  is the sample size. The reported minimum region volume is  $\min_k \hat{V}_k$ . The mean score margin is

$$\frac{1}{N} \sum_{n=1}^N (s_{(1)}(\mathbf{z}_n) - s_{(2)}(\mathbf{z}_n)), \tag{21}$$

where  $s_{(1)}$  and  $s_{(2)}$  denote the largest and second-largest affine scores, respectively.

Targeted sampling is used only to check that each analytically constructed winning interval is attainable. For an open winning interval  $(a_k, b_k)$ , choose any  $\rho_k^* \in (a_k, b_k)$ , set  $u_k = 1/(1 + \rho_k^*)$ , and let  $\mathbf{z}_k^* = (1 - u_k)\mathbf{p}_1 + u_k\mathbf{p}_2$ . Then  $\rho(\mathbf{z}_k^*) = \rho_k^*$  exactly. Targeted sampling therefore checks the non-emptiness of the constructed winning intervals; it is not used to estimate their probability mass or volume.

The Monte Carlo quantities are stochastic estimates, so the final decimal digits may vary with the pseudorandom-number generator state. The reported comparisons use the same sampling domain, prototype locations, class and dimension grids, sampling budget, soft labels, and score definitions within each corresponding configuration.